

AI-Exposed Earnings and Infrastructure Shocks: High-Frequency Evidence on Markets and Macroeconomic Expectations

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ECON 31730 Advanced Time Series Analysis

Abstract

This paper asks whether market reactions to AI-exposed earnings and infrastructure news predict later macro-financial and firm-level outcomes. I construct a high-frequency event-shock proxy from intraday abnormal returns of an AI infrastructure basket around earnings releases and major AI model, infrastructure, and regulatory announcements. The strongest evidence is reduced form: positive event shocks forecast equity-market responses and higher capex responses among more AI-exposed firms, while aggregate real-activity responses are mixed. Because the external-instrument first stage is borderline rather than strong, conditional causal evidence is reported through Anderson–Rubin weak-IV sets and LP-IV relative responses normalized by the monthly AI-basket abnormal return. The evidence supports market-based AI repricing, but not precise structural impulse responses without stronger shock relevance and additional recoverability assumptions.

1 Introduction

Financial markets are often the first place where investors aggregate news about AI demand, infrastructure scarcity, and frontier-model progress. The empirical question is whether those market reactions contain information about later macro-financial and firm-level outcomes. The central difficulty is that AI news is not an observed structural shock. Earnings announcements, model releases, export-control decisions, and infrastructure partnerships can all mix expected cash-flow news, discount-rate news, supply constraints, and broad market sentiment.

This paper studies that problem with a high-frequency event design. The baseline shock is an AI-exposed earnings and infrastructure event-shock proxy: the intraday abnormal return of an AI infrastructure basket around 116 Tier 1 earnings announcements and 19 major Tier 2 AI events. The event proxy is used first as a reduced-form predictor and then, more cautiously, as an external instrument for the monthly AI-basket abnormal return. A companion fixed-income event study follows the logic of Andrews and Farboodi (2025): pure model releases can be informative about transformative-AI beliefs if they move long-horizon rates.

The main empirical contribution is deliberately disciplined. Reduced-form local projections ask whether the event-shock proxy predicts later outcomes. Weak-IV Anderson–Rubin sets then

describe what the same evidence permits under the external-instrument interpretation. Conventional LP-IV coefficients are shown only as relative impulse responses normalized by the impact movement in the monthly AI-basket abnormal return. The paper also reports firm-level exposure interactions, frequency-domain diagnostics, BVAR and proxy-SVAR checks, sign restrictions, and state-space extensions, but none of those tools is treated as a substitute for shock validity or first-stage relevance.

2 Empirical Framework

All time-series objects below are defined for the transformed outcomes in Table 1; the SVMA, Wold, and spectral arguments require working with stationary growth rates, differences, or return objects rather than raw trending levels. Let the macro-financial state have a structural moving-average representation

$$y_t = \Theta(L)\varepsilon_t,$$

where the AI-related innovation is one component of ε_t . Second moments of y_t alone do not identify that component, so the paper uses an external instrument. The maintained proxy equation is

$$z_t = \alpha \varepsilon_{AI,t} + v_t,$$

with relevance for the AI component and exclusion from the other structural innovations. Under those restrictions, the instrument identifies relative impulse responses up to a normalization (Stock and Watson, 2018). Stronger objects, such as an absolute shock series, historical decompositions, or conventional FEVDs, require additional scale and recoverability assumptions.

Local projections provide the baseline time-series estimator (Jorda, 2005). In population, sufficiently controlled LPs and VARs estimate the same impulse responses (Plagborg-Møller and Wolf, 2021). In this application the first stage is borderline, so the estimator hierarchy is conservative: reduced-form LPs first, Anderson–Rubin weak-IV sets second, conditional LP-IV relative responses third, and VAR/SVAR or Bayesian estimates only as triangulation.

The monthly outcomes are persistent, so the baseline uses lag-augmented local projections with Eicker–Huber–White standard errors following Montiel Olea and Plagborg-Møller (2021). Newey–West, fixed- b , bootstrap, and panel-clustered standard errors are reported as robustness checks where their assumptions are more appropriate. Frequency-domain evidence is reported through spectra, coherency, and band shares. These are diagnostic measures of association and cycle length; they are not labeled as structural FEVDs unless a full structural decomposition is imposed.

The explicit mapping from these methods to the course notes and problem sets is kept in Appendix B. The main text uses the method names directly so that the empirical claims can be read without the course-specific map.

3 Data and Events

The baseline event universe has 135 events and uses off-hours events mapped to the next regular trading session open. Tier 1 events are earnings announcements for NVDA, MSFT, GOOGL, META, AMZN, AVGO, ORCL, AMD, and TSM. Tier 2 events are major model-release, regulatory, partnership, and AI infrastructure announcements. WRDS TAQ minute data currently covers 78982 event-ticker-minute observations.

The AI infrastructure basket contains semiconductors, semiconductor equipment, networking, and data-center power/cooling firms. QQQ and SPY are benchmark assets used for market adjustment and robustness, not members of the AI basket B_e . The event database keeps the original event timestamp, the effective regular-session timestamp, timestamp source, primary tickers, LSEG headline evidence where available, and the off-hours rule.

The timestamp rule is hybrid. For the main 2022–2026 baseline, event time is the earliest audited market-observable LSEG/Refinitiv headline or earnings-release timestamp that directly conveys the event. If the earliest market-observable timestamp is a pre-announcement, social-wire, or non-official source, the event remains in the all-events baseline only with an explicit caveat and timing/drop sensitivity. For the 2017–2022 extension, only LSEG-reviewed Tier 2 timestamps enter the preferred extension; public/newswire fallback timestamps are appendix-only.

Macro data come from FRED baseline series, FRED-MD/QD, Fernald utilization-adjusted TFP, BEA fixed assets, BLS QCEW, EIA electricity, Compustat, SEC EDGAR capex facts, and ESZ firm-level AI exposure scores. Firm-level exposure regressions use a broad quarterly panel combining CRSP, Compustat, ESZ, and I/B/E/S. Private data currently used are WRDS TAQ, CRSP, Compustat, and I/B/E/S; LSEG is used for news timestamp validation and daily prices.

Table 1: Outcome transformations and units. The event shock z_t and normalization n_t are decimal returns. For any event-shock coefficient in the main tables, a one-percentage-point event shock corresponds to 0.01 times the displayed coefficient.

Series	Transformation	Paper units	Role
Industrial production, payrolls, real PCE, S&P 500	100 times log change	monthly percentage-point log change	stationary growth-rate outcome
PCE and CPI price indexes	1200 times log change	annualized monthly inflation rate	stationary inflation outcome
Treasury yields, unemployment, Fed funds, VIX	monthly first difference	monthly level change	persistent level series used in differences
Event shock z_t and normalization n_t	log-return abnormal returns	decimal return units	1 percentage point shock equals 0.01 in tables

4 Shock Construction

For event e , ticker j , and event window w , define the log return

$$r_{j,e}^{(w)} = \log P_{j,e,end}^{(w)} - \log P_{j,e,start}^{(w)}.$$

Let $B_e = \{j \in \mathcal{B}_{AI} : P_{j,e,start} \text{ and } P_{j,e,end} \text{ are observed}\}$. The AI basket return is the equal-weight return over these available AI-basket constituents:

$$r_{B,e}^{(w)} = \sum_{j \in B_e} \omega_{j,e} r_{j,e}^{(w)}, \quad \omega_{j,e} = \frac{1}{|B_e|}.$$

The market-adjusted event shock is

$$z_e^{(w)} = r_{B,e}^{(w)} - \hat{\beta}_B r_{M,e}^{(w)}.$$

The baseline uses QQQ as M , with SPY as a robustness benchmark. The coefficient $\hat{\beta}_B$ is estimated from daily AI-basket and benchmark returns outside the event-window calculation.

The monthly instrument is

$$z_t^{eq} = \sum_{e \in t} z_e^{(w)}.$$

The normalization variable is the monthly AI-basket abnormal return

$$n_t = r_{AI,t}^{month} - \hat{\beta}_M r_{M,t}^{month}.$$

This separates the high-frequency instrument from the monthly impulse being instrumented.

The implemented baseline is the pre-registered post-60-minute, QQQ-adjusted, event-excluded-beta, all-events equity shock rather than a specification selected by first-stage strength. Because 135 current-sample events are dominated by Tier 1 earnings announcements, the paper refers to this object as an AI-exposed earnings and infrastructure event-shock proxy, not as a pure AI capex structural shock. The selected specification has 135 finite event-window shocks. There are no missing selected-shock observations in the current baseline; the construction report still records missing-reason fields for non-selected variants and future data pulls, so the paper does not silently condition on successful data pulls. WRDS TAQ remains the preferred source when available; Databento U.S. equities minute data fill missing AI-basket and benchmark coverage for post-2023 events. For Tier 2 events, the shock uses the hybrid timestamp rule from the data section: first audited market-observable LSEG/Refinitiv tape in the current baseline, caveated sensitivity for pre-announcement/social/non-official tape, and appendix-only treatment for public/newswire fallback Tier 2 timestamps in the 2017–2022 extension.

The construction code also reports robustness variants by event window, benchmark, sample filter, and beta-estimation rule. These variants are used as sensitivity checks, not as the main identifying design.

5 Identification

5.1 LP-IV Scaling

Following the external-instrument setup in Stock and Watson (2018), impose relevance and exclusion:

$$E[z_t \varepsilon_{AI,t}] \neq 0, \quad E[z_t \varepsilon_{-AI,t}] = 0.$$

For horizon h , let

$$S_h = \{t : y_{i,t+h}, n_t, z_t, X_{t-1}^{(p)} \text{ are observed}\}$$

be the available LP sample. Let X_h stack a constant and the predetermined controls $X_{t-1}^{(p)}$ on that sample, and define

$$M_{X,h} = I - X_h(X_h' X_h)^{-1} X_h'.$$

The residualized variables are horizon-specific:

$$\tilde{z}_{t,h} = M_{X,h} z_t, \quad \tilde{n}_{t,h} = M_{X,h} n_t, \quad \tilde{y}_{i,t+h} = M_{X,h} y_{i,t+h}.$$

The just-identified LP-IV estimand is then

$$\hat{\beta}_{i,h}^{IV} = \frac{\sum_{t \in S_h} \tilde{z}_{t,h} \tilde{y}_{i,t+h}}{\sum_{t \in S_h} \tilde{z}_{t,h} \tilde{n}_{t,h}}.$$

Under the proxy restrictions and the maintained controlled-information-set projection,

$$E[\tilde{z}_{t,h} \tilde{y}_{i,t+h}] = \alpha \Theta_{i,h}^{AI}, \quad E[\tilde{z}_{t,h} \tilde{n}_{t,h}] = \alpha \Theta_{n,0}^{AI},$$

so

$$\beta_{i,h}^{IV} = \frac{\Theta_{i,h}^{AI}}{\Theta_{n,0}^{AI}}.$$

The response is normalized to a one-unit impact movement in n_t , not to a one-standard-deviation structural shock unless an additional scale assumption is imposed.

5.2 Recoverability and Invertibility

Let x_t denote a target state component and u_t denote the Wold innovations in the observed macro-financial information set. The projection result used in the recoverability appendix shows that future Wold innovations add no linear information for x_t exactly when

$$\text{Cov}(x_t, u_{t+m}) = 0 \quad \forall m \geq 1.$$

Mapped to the proxy-SVMA setting, recoverability is weaker than invertibility and is needed for stronger objects such as an absolute shock series. The testable implication of invertibility used here

is that lagged z_t should not Granger-cause y_t after conditioning on lagged y_t . This is a diagnostic implication, not a proof that the AI innovation is recoverable.

5.3 Multiple-Instrument GMM

When candidate instruments such as the equity shock, fixed-income belief shock, and curated news shock are used together, the moment condition is

$$E \left[\tilde{Z}_t (\tilde{y}_{i,t+h} - \beta_h \tilde{n}_t) \right] = 0.$$

The efficient-GMM estimator and Hansen J -test use standard linear GMM logic. Rejection of the J -test is interpreted as evidence against the joint overidentifying restrictions across the candidate instruments, not as automatic rejection of the baseline equity instrument.

6 Inference

The baseline LP is lag-augmented:

$$y_{i,t+h} = \alpha_h + \beta_h n_t + \Gamma'_h X_{t-1}^{(p)} + u_{i,t+h},$$

where $X_{t-1}^{(p)}$ contains only predetermined lags of n_t , z_t , event counts, and macro outcomes. The lag-augmentation is written this way so current-month post-event information is not used as a control. The result in Montiel Olea and Plagborg-Moller (2021) implies that Eicker-White standard errors are valid under the stated mean-independence, known-lag-length, nonsingularity, moment, and persistence conditions, including local persistence and horizons satisfying $h/T \rightarrow 0$. Lag-order sensitivity is reported separately because post-selection lag choice would otherwise add unmodeled inference uncertainty.

For weak instruments, the paper reports Anderson-Rubin intervals based on testing

$$H_0 : \beta_h = \beta_0$$

in the auxiliary regression

$$y_{i,t+h} - \beta_0 n_t = \alpha_h + \Gamma'_h X_{t-1}^{(p)} + \delta_h z_t + u_{i,t+h}.$$

The null is accepted when the robust test of $\delta_h = 0$ does not reject. The reported sets explicitly distinguish bounded, lower-unbounded, upper-unbounded, grid-unbounded, disjoint, and empty cases. Current generated counts are 174 bounded, 1 unbounded, 0 disjoint, and 0 empty sets. For VAR/SVAR estimates, uncertainty is assessed with recursive residual bootstrap. For panel regressions, dependence is handled with firm clustering and block bootstrap checks.

The current baseline is materially improved after filling missing basket coverage, but it still does not clear strong-IV thresholds uniformly. The robust first-stage statistic is 14.994 with partial

$R^2 = 0.1394$. The 23.1 cutoff is used as a conservative Montiel Olea–Pflueger effective- F screening rule for the paper’s high-relevance tolerance (Montiel Olea and Pflueger, 2013); it is not a universal Stock–Yogo-style guarantee that conventional 2SLS inference is reliable in every horizon-specific sample. Consequently, conventional 2SLS confidence intervals are reported as descriptive objects except where horizon-specific first stages are strong, while Anderson–Rubin confidence sets and reduced-form responses remain the primary frequentist evidence.

The paper distinguishes three types of uncertainty: sampling uncertainty around point-identified conditional LP-IV IRFs, weak-IV robust set uncertainty from Anderson–Rubin inversion, and prior-weighted posterior summaries under BVAR/sign-restriction Bayesian implementations.

7 Baseline Results

The empirical results are ordered by the strength of the identification evidence. The baseline sample has 135/ 135 finite event shocks, and the main first-stage robust F -statistic is 14.994 with partial $R^2 = 0.1394$. This clears the conventional weak-IV warning threshold of 10, but it does not clear the stronger 23.1 threshold used as the paper’s high-relevance gate. The main results therefore lead with reduced-form local projections and Anderson–Rubin confidence sets. Conventional LP-IV point estimates are shown only afterward, as conditional relative responses normalized by the monthly AI-basket abnormal return.

The validity diagnostics in Table 2 explain the ordering. The timing gate passes, but the event proxy is not a laboratory shock: 92 of 135 events are confound-flagged, 66 event windows use mixed data sources, and the macro-predictability diagnostic remains active. These facts do not invalidate reduced-form event-shock LPs, but they discipline the causal language. They also do not prove the external-instrument exclusion restriction; they are plausibility checks for the maintained proxy-IV assumptions.

7.1 Reduced-Form Event-Shock Responses

The first object is the reduced-form local projection

$$y_{i,t+h} = \alpha_{i,h} + \theta_{i,h}z_t + \Gamma'_{i,h}X_{t-1}^{(p)} + e_{i,t+h}.$$

These coefficients are responses to the observed high-frequency event-shock proxy z_t . They do not divide by the first stage and therefore do not require the stronger normalization needed to interpret LP-IV coefficients as scaled structural responses. The table reports the core outcomes at horizons 0, 6, 12, and 24 months. For transformed growth outcomes, these are horizon- h growth-rate responses, not cumulative level responses. The reduced form is intentionally conservative: it answers whether the market-observed AI event shock predicts later outcomes, not whether the shock has been fully scaled into an economically primitive AI innovation.

The core reduced-form estimates are mixed for real activity and stronger for equity returns.

Table 2: Shock validity diagnostics for the selected current-sample event-shock proxy. Timing passes, but confounds, source mixing, macro predictability, and near-warning whiteness diagnostics require reduced-form and weak-IV robust evidence to lead.

Diagnostic	Value	Interpretation
Event shocks	135/135	complete selected-shock coverage
Missing timestamps	0	timing gate passes
Duplicate effective timestamps	36	cluster-collapse sensitivity required
Next-session mappings	122	off-hours announcements mostly repriced at regular open
Confound-flagged events	92/135	large confound burden; drop-confound variant reported
Mixed-source windows	66	source-mixing flags remain visible
Ljung-Box p-value, lag 6	0.064	near the warning region, not a clean white-noise pass
ARCH-LM p-value, lag 6	0.085	heteroskedasticity warning is mild but relevant
Macro predictability R2	0.289	predictability warning; residualized variant reported
Periodogram flatness CV	0.729	shock spectrum is diagnostic, not assumed flat
Invertibility Granger rejections	0/8	no broad rejection at 10 percent in generated diagnostics

Table 3: Reduced-form local projection responses to the AI event-shock proxy. Coefficients are horizon- h responses to z_t , using the same controls and sample as the LP-IV specification. Growth outcomes are not cumulated. The shock is measured in decimal return units; a one-percentage-point event shock corresponds to 0.01 times the displayed coefficient.

Outcome	Horizon	RF coef.	RF 90% lower	RF 90% upper	N
Industrial production	0	2.307	-10.814	15.429	100
Industrial production	6	-14.650	-28.846	-0.454	99
Industrial production	12	16.249	-4.706	37.204	93
Industrial production	24	18.780	-43.249	80.808	81
Payroll employment	0	4.258	-7.092	15.607	100
Payroll employment	6	-7.888	-19.766	3.990	100
Payroll employment	12	14.717	-3.312	32.747	94
Payroll employment	24	-82.166	-144.907	-19.425	82
S&P 500	0	9.221	-34.142	52.584	100
S&P 500	6	55.719	11.579	99.859	100
S&P 500	12	59.727	15.500	103.953	95
S&P 500	24	217.940	120.512	315.368	83

Industrial production is negative at six months, positive but imprecise by twelve months, and very imprecise at twenty-four months. Payroll employment is negative at the twenty-four-month horizon in this specification. The S&P 500 response is positive and statistically separated from zero at the six-, twelve-, and twenty-four-month horizons. These patterns should be read as predictive responses to the event-shock proxy rather than as fully scaled structural impulse responses.

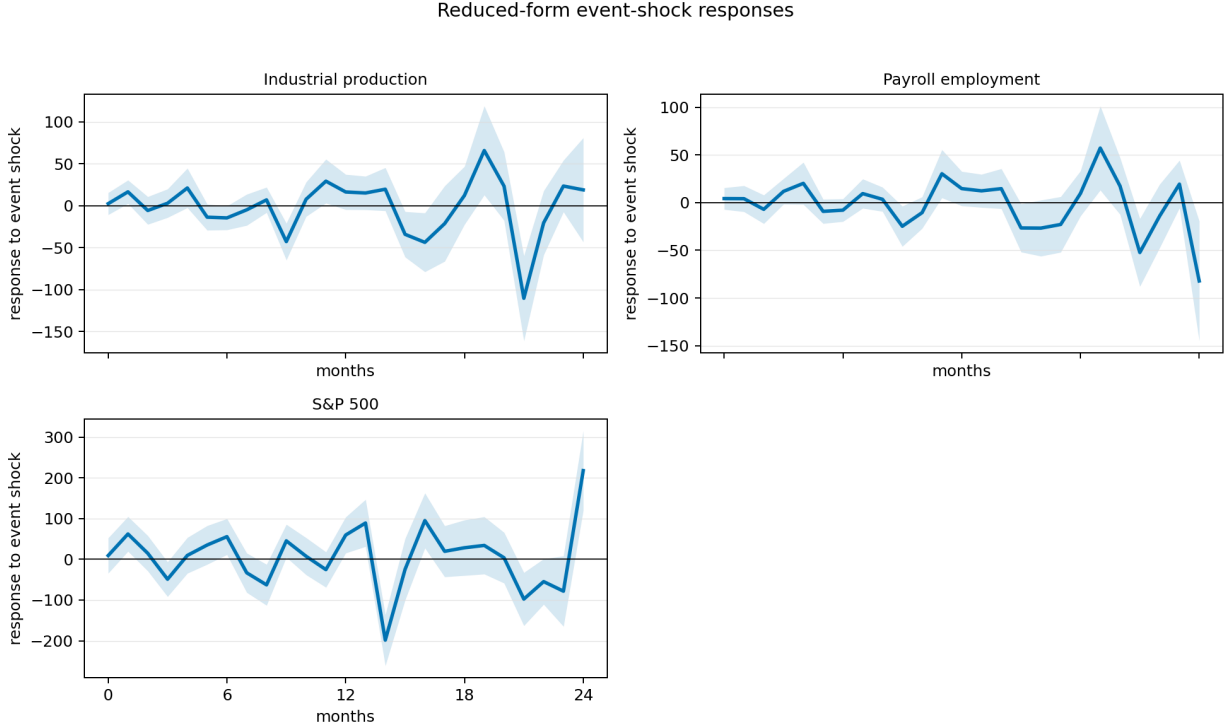


Figure 1: Reduced-form responses for the core outcomes. The line is the horizon- h response to the event-shock proxy z_t ; the shaded band is the 90 percent EHW interval. Growth outcomes are shown as growth-rate responses, not cumulative level effects.

7.2 Weak-IV Evidence

The horizon-specific first-stage table makes the weak-IV status visible next to the response horizons. The relevant conclusion is not that the instrument is irrelevant. The coefficient has the expected positive sign and the baseline first stage is borderline. The issue is that first-stage strength varies by horizon because the available sample changes as h increases, so conventional LP-IV standard errors alone would understate the uncertainty in the longer-run responses.

Table 6 reports weak-IV robust confidence sets for the same core outcomes and horizons. Across the full baseline grid there are 174 bounded sets and 1 unbounded or grid-unbounded sets. The displayed core outcomes have bounded sets at the reported horizons, but some long-horizon first stages fall below 10. This is why the AR sets, rather than conventional LP-IV standard errors, are the main weak-IV uncertainty summary for the conditional LP-IV interpretation.

Table 4: First-stage reconciliation across diagnostics. The robust F -statistics differ because the rows use different residualization schemes, samples, instruments, and purposes. The main LP-IV row is the relevant statistic for the baseline conditional LP-IV table; the other rows are diagnostics or robustness gates.

Statistic	Instrument	Sample	Robust F	Partial R2	Purpose
Main LP-IV first stage	$z_{eq_selected}$	2022–2026 LP sample	14.994	0.139	baseline conditional LP-IV scaling
Shock-validity diagnostic first stage	$z_{eq_selected}$	selected-shock monthly panel	2.261	0.029	diagnostic only; not the main LP-IV table
Multi-IV GMM effective F	equity, residualized, drop-confound, drop-announcer, FI	2022–2026 LP sample	7.750	0.273	overidentified robustness
Proxy-SVAR residual first stage	$z_{eq_selected}$	VAR residual sample	1.909	n/a	stronger-assumption SVAR cross-check
2017–2022 extension first stage	extension $z_{eq_selected}$	WRDS TAQ extension	4.611	0.038	appendix coverage/relevance gate
Pre-registered capex/infrastructure-SPY candidate	capex/infrastructure, SPY	2022–2026 candidate	20.443	0.177	stronger-shock candidate, still below strong-IV gate

Table 5: Horizon-specific first-stage diagnostics for core outcomes. The robust F -statistic and partial R^2 are reported on the exact horizon-specific sample used for each local projection.

Outcome	Horizon	Robust F	Robust t	Partial R2	First-stage coef.	N
Industrial production	0	14.994	3.872	0.139	1.182	100
Industrial production	6	15.296	3.911	0.155	1.266	99
Industrial production	12	16.393	4.049	0.187	1.451	93
Industrial production	24	3.779	1.944	0.075	1.473	81
Payroll employment	0	14.994	3.872	0.139	1.182	100
Payroll employment	6	14.994	3.872	0.139	1.182	100
Payroll employment	12	17.583	4.193	0.198	1.499	94
Payroll employment	24	9.537	3.088	0.160	2.021	82
S&P 500	0	14.994	3.872	0.139	1.182	100
S&P 500	6	14.994	3.872	0.139	1.182	100
S&P 500	12	16.215	4.027	0.175	1.310	95
S&P 500	24	11.377	3.373	0.187	2.058	83

Table 6: Weak-IV robust Anderson–Rubin 90 percent confidence sets. Set status is reported explicitly because weak instruments can produce bounded, unbounded, disjoint, or empty sets. Coefficients use the same normalized units as the LP-IV table; for a one-percentage-point movement in n_t , multiply endpoints by 0.01.

Outcome	Horizon	AR 90% lower	AR 90% upper	AR set	Robust F
Industrial production	0	-11.144	13.713	bounded	14.994
Industrial production	6	-26.449	-0.351	bounded	15.296
Industrial production	12	-4.221	26.952	bounded	16.393
Industrial production	24	-158.451	53.771	bounded	3.779
Payroll employment	0	-7.163	13.911	bounded	14.994
Payroll employment	6	-18.533	3.974	bounded	14.994
Payroll employment	12	-2.814	23.018	bounded	17.583
Payroll employment	24	-138.157	-7.021	bounded	9.537
S&P 500	0	-30.284	54.957	bounded	14.994
S&P 500	6	8.942	110.823	bounded	14.994
S&P 500	12	10.982	104.902	bounded	16.215
S&P 500	24	54.641	215.960	bounded	11.377

7.3 Conditional LP-IV Relative IRFs

The LP-IV estimand is

$$\beta_h^{IV} = \frac{\sum_{t \in S_h} \tilde{z}_{t,h} \tilde{y}_{t+h}}{\sum_{t \in S_h} \tilde{z}_{t,h} \tilde{n}_{t,h}},$$

so it is a relative response normalized by the monthly AI-basket abnormal return n_t . These estimates are useful for comparing signs and magnitudes on the normalized scale, but they are not the headline evidence because the first stage is borderline rather than strong. They should be interpreted as relative IRFs conditional on the external-instrument assumptions and on the chosen normalization, not as one-standard-deviation structural AI shocks.

7.4 Firm-Level Exposure Responses

The second empirical pillar uses the broad CRSP/Compustat/ESZ/IBES quarterly panel to ask whether firms with higher ex-ante AI exposure react differently to the same aggregate AI event-shock proxy. Let ξ_f denote the standardized Eisfeldt–Schubert–Zhang–Taska AI-exposure score (Eisfeldt et al., 2023). The reduced-form interaction specification is

$$y_{f,q+h} = \alpha_f + \tau_q + \theta_h(z_q \xi_f) + e_{f,q+h},$$

with firm and quarter fixed effects. Quarter effects absorb the aggregate shock level, so θ_h is identified only from differential responses across firms with different ξ_f in quarters with different shock

Table 7: Conditional LP-IV relative impulse responses. Estimates instrument the monthly AI-basket abnormal return n_t with the event-shock proxy z_t ; inference should be read alongside the Anderson–Rubin sets in Table 6. Both z_t and n_t are decimal return objects, so these are relative responses normalized by a one-unit movement in n_t . For a one-percentage-point movement in n_t , multiply coefficients by 0.01.

Outcome	Horizon	LP-IV coef.	EHW s.e.	NW s.e.	Robust F	IV category	N
Industrial production	0	1.952	6.682	6.989	14.994	borderline	100
Industrial production	6	-11.576	7.016	7.402	15.296	borderline	99
Industrial production	12	11.198	8.380	9.002	16.393	borderline	93
Industrial production	24	12.753	22.292	20.228	3.779	weak	81
Payroll employment	0	3.603	5.727	6.165	14.994	borderline	100
Payroll employment	6	-6.674	6.050	6.603	14.994	borderline	100
Payroll employment	12	9.821	7.020	7.133	17.583	borderline	94
Payroll employment	24	-40.646	28.020	27.541	9.537	weak	82
S&P 500	0	7.802	22.670	24.080	14.994	borderline	100
S&P 500	6	47.147	26.532	24.361	14.994	borderline	100
S&P 500	12	45.584	24.716	20.605	16.215	borderline	95
S&P 500	24	105.902	37.691	35.254	11.377	borderline	83

realizations. The companion interaction-IV specification replaces $z_t\xi_f$ with $n_q\xi_f$ and instruments it with $z_q\xi_f$. Both regressors are scaled as 100 times the shock times a one-standard-deviation exposure score, so coefficients are outcome percentage-point responses to a one-percentage-point aggregate shock for a one-standard-deviation more AI-exposed firm. The corresponding first stage is

$$n_q\xi_f = \alpha_f + \tau_q + \pi_h(z_q\xi_f) + v_{f,q}.$$

Table 8: Firm-level exposure interaction results. The reduced-form columns are the main evidence in this table. Interaction-IV columns instrument $n_q\xi_f$ with $z_q\xi_f$ and are reported as weak-IV companion estimates. All specifications include firm and quarter fixed effects and report two-way clustered standard errors.

Firm outcome	Horizon (qtrs.)	RF coef.	RF two-way s.e.	Interaction-IV coef.	IV two-way s.e.	First-stage F	Partial R2	N	Firms	Quarters	Event quarters
Capex/assets	4	0.010	0.003	0.017	0.019	1.057	0.021	74396	3700	24	7
IBES EPS surprise	1	4.352	2.517	11.255	10.405	1.613	0.015	5508	225	35	12
IBES revision balance	1	0.138	0.354	0.352	0.871	1.249	0.017	5727	230	35	12
Operating margin	4	0.145	0.107	0.470	0.369	0.411	0.006	91945	3511	32	9
Quarterly return	0	-0.092	0.059	-0.293	0.296	0.427	0.006	100112	3551	32	9
Quarterly return	1	-0.057	0.084	-0.316	0.888	0.132	0.002	96638	3551	31	8
R&D/assets	4	0.001	0.002	0.003	0.010	0.430	0.006	44268	1947	32	9
Sales growth	4	0.029	0.048	0.093	0.108	0.404	0.006	99006	3694	32	9

The broad panel contains up to 100112 firm-quarter observations and 3700 firms in the displayed specifications. The cleanest positive reduced-form exposure result is capex, which is estimated from 7 event quarters in the displayed row: a one-percentage-point event shock is associated with a 0.010

percentage-point higher capex-to-assets response four quarters later for a one-standard-deviation more AI-exposed firm. The interaction-IV first stage is weak, with a maximum displayed two-way clustered first-stage F -statistic of 1.613. The cross-section is therefore kept in the main paper as reduced-form exposure evidence rather than as a separately strong causal IV design. The result still adds economic content: the aggregate event shock predicts where real investment responds most, even when the time-series normalization remains too weak for conventional LP-IV point estimates to lead.

8 Frequency-Domain Evidence

The spectrum of a covariance-stationary outcome satisfies

$$\text{Var}(y_t) = \int_{-\pi}^{\pi} s_y(\omega) d\omega.$$

The convention is

$$s_y(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_y(k) e^{-i\omega k}.$$

Thus $s_y(\omega)d\omega$ is interpretable as the variance contribution of cycles with frequencies in $[\omega, \omega + d\omega]$.

The baseline reports squared coherency:

$$|K_{yz}(\omega)|^2 = \frac{|s_{yz}(\omega)|^2}{s_y(\omega)s_z(\omega)}.$$

Equivalently, this equals $s_{y^\dagger}(\omega)/s_y(\omega)$, where $y^\dagger$ is the best linear projection of y on all leads and lags of z . It is therefore a frequency-domain signal-strength measure.

The frequency-domain evidence is summarized over high-frequency cycles shorter than 6 months, business-cycle frequencies between 6 and 32 months, and low frequencies longer than 32 months. These outputs are labeled coherencies, band shares, or IV-linked spectral associations in the baseline. They are not conventional FEVDs unless the full structural decomposition assumptions are imposed.

9 Robustness and Extensions

9.1 Shock-Construction Variants

The first robustness layer changes the shock construction while preserving the pre-registered baseline. These variants are robustness screens, not an automatic model-selection rule. The goal is to learn whether the baseline result is driven by confounded dates, by Tier 1 earnings, by Tier 2 narrative events, by regulatory news, or by the benchmark used in abnormal-return construction.

The sweep has three useful implications. First, the economically preferred baseline remains borderline rather than strong: the all-events specification has robust $F = 14.994$. Second, the

Table 9: Shock-construction robustness. The table compares first-stage strength and weak-IV set behavior across pre-specified variants of the current-sample event-shock proxy.

Variant	Robust F	Partial R2	Macro pred. R2	Bounded AR sets	Grid-unbounded AR sets	First-stage gate
All events baseline	14.994	0.139	0.295	12	0	borderline
Residualized shock	12.537	0.108	0.124	11	1	borderline
Drop confound events	15.428	0.125	0.273	11	1	borderline
Tier 1 earnings only	15.267	0.145	0.261	11	1	borderline
Tier 2 events only	0.518	0.007	0.234	1	11	weak
Capex/infrastructure only	15.925	0.155	0.273	12	0	borderline
Model releases only	0.376	0.002	0.353	1	11	weak
No regulation events	15.131	0.140	0.295	12	0	borderline
QQQ benchmark baseline	14.994	0.139	0.295	12	0	borderline
SPY benchmark variant	15.921	0.128	0.254	12	0	borderline

residualized shock substantially lowers the macro-predictability diagnostic, from about 0.295 to about 0.124, but it does not turn the first stage into a strong-IV design. Third, Tier 2-only and model-release-only variants are weak, while Tier 1-only, capex/infrastructure-only, no-regulation, and SPY-benchmark variants remain borderline. This supports the paper’s choice to keep the full baseline for power and transparency, while reporting narrowed variants as robustness checks.

9.2 Event Clustering and Mechanical Overlap

Several current-sample announcements share the same effective regular-session timestamp or effective event date. Since the monthly instrument sums event shocks, overlapping windows can otherwise double-count the same market move. The first sensitivity collapses duplicate effective timestamps by averaging shocks within a timestamp cluster before monthly aggregation; a stricter appendix check does the same at the effective-date level.

Table 10: Cluster-collapsed shock sensitivities. Duplicate event-window shocks are averaged within timestamp or date clusters before monthly aggregation. The baseline conclusion is unchanged: the first stage remains borderline rather than strong.

Specification	Instrument	Nonzero months	Robust F	Partial R2	IV category	Bounded AR sets	Unbounded/grid AR sets	Paper use
Timestamp-cluster collapsed	z_eq_cluster_timestamp	39	12.814	0.120	borderline	174	1	cluster sensitivity
Date-cluster collapsed	z_eq_cluster_date	39	14.409	0.120	borderline	174	1	stricter appendix cluster sensitivity

The second sensitivity asks how much of the LP-IV first stage comes from the fact that event-window returns are mechanically part of the monthly AI-basket abnormal return n_t . It subtracts the event-window shock from n_t and reruns the same LP-IV stack. The first stage falls sharply, so the paper treats LP-IV primarily as a conditional rescaling of the reduced form rather than as the leading source of evidence.

Table 11: Mechanical-overlap sensitivities. The endogenous monthly normalization variable is adjusted by subtracting the event-window shock. The resulting weak first stages reinforce the reduced-form-first interpretation.

Specification	Endogenous variable	Instrument	Robust F	Partial R2	IV category	Bounded AR sets	Unbounded/grid AR sets	Paper use
Baseline n minus event-window shock	n_ai_monthly_ex_event	z_eq_selected	0.355	0.004	weak	16	159	mechanical-overlap sensitivity
Cluster n minus cluster shock	n_ai_monthly_ex_event_cluster	z_eq_cluster_timestamp	0.039	0.000	weak	17	158	cluster plus mechanical-overlap sensitivity

9.3 Pre-Registered Stronger-Shock Candidate

Because the baseline first stage is borderline, the most valuable next identification improvement is a better shock, not another estimator. The paper therefore defines a small candidate set using economic restrictions before looking at first-stage strength. The leading candidate narrows the shock to Tier 1 earnings plus capex/infrastructure partnership events and uses SPY rather than QQQ as the market adjustment. This choice is motivated by the channel: the candidate targets broad AI-infrastructure repricing while removing Nasdaq-sector comovement.

Table 12: Pre-registered stronger-shock candidates. The candidate set is fixed by economic restrictions rather than by searching over every possible first-stage statistic.

Candidate	Robust F	Partial R2	Macro pred. R2	Nonzero months	First-stage gate	Paper use
Current baseline	14.994	0.139	0.295	39	borderline	improves relevance but AR/reduced-form still lead
Capex/infrastructure, SPY	20.443	0.177	0.239	39	borderline	improves relevance but AR/reduced-form still lead
Capex/infrastructure, QQQ	16.031	0.154	0.248	39	borderline	improves relevance but AR/reduced-form still lead
Tier 1 earnings, QQQ	11.811	0.119	0.230	39	borderline	improves relevance but AR/reduced-form still lead
Drop confounds, SPY	0.058	0.001	0.297	31	weak	appendix or descriptive only
No regulation, SPY	12.061	0.107	0.243	39	borderline	improves relevance but AR/reduced-form still lead

The leading capex/infrastructure-SPY candidate raises the robust first-stage F -statistic to 20.443, which is a meaningful improvement over the current baseline but still below the strong-IV threshold. It should therefore be treated as the next pre-registered candidate and a robustness design, not as a reason to make conventional LP-IV point estimates the primary result.

9.4 Capex-Guidance Surprise Candidate

The next tightening step uses more information from the earnings tape. The analysis builds capex/cloud-guidance surprise candidates from LSEG/Refinitiv earnings headlines and reviewed story text, then adds WRDS IBES summary-history revisions for the same target firms. A row enters the primary text-based candidate only if the event is an earnings or capex-partnership event and the LSEG tape contains both guidance language and AI-infrastructure, cloud, compute, or AI-demand language. The IBES revisions are used as diagnostics because analyst revisions occur after the high-frequency event window. The detailed candidate grid is retained in the generated artifacts rather than as a main-text table.

This exercise is informative but not a better first-stage design. The WRDS target-firm IBES

pull adds 8393 EPS summary-history rows and the text screen identifies 41 candidate event windows for the strongest capex-guidance specification. The best candidate, Revision-weighted guidance, SPY, has robust $F = 1.221$. That is below the borderline threshold of 10 and far below the strong-IV threshold. The result rules out a tempting but too-narrow LSEG text filter as the immediate solution. Stronger causal evidence would require true capex-guidance expectation surprises from StreetAccount/LSEG guidance fields or analyst capex/cloud-revenue forecasts, not additional post-event text filters.

9.5 Structured Guidance-Surprise Shock

The strongest route to sharper causal claims is a shock that measures the surprise in capex or cloud guidance relative to a pre-event expectation. The robustness section therefore adds a structured WRDS/IBES guidance screen. The input is IBES guidance detail data joined to the IBES identifier file, not post-event EPS revision histories. For each guidance row, the company guidance midpoint is compared with the IBES `mean_at_date` consensus available at the announcement:

$$s_e = \frac{g_e - \bar{g}_{e,-}}{|\bar{g}_{e,-}| + \epsilon}.$$

The primary candidate uses capital-expenditure guidance (**CPX**); a secondary candidate adds sales guidance only when the LSEG tape confirms an AI/cloud/infrastructure context; and an EPS-only guidance shock is reported as a non-capex placebo. The detailed candidate grid is retained as a generated artifact; the main text reports only the gate result.

The structured guidance pull adds 762 target-firm guidance rows and identifies 36 nonzero event windows for the strongest displayed candidate. The best candidate, IBES CPX plus AI/cloud sales surprise, has robust $F = 3.464$, so the gate is weak. Structured guidance is therefore useful measurement evidence, but it does not overturn the weak-IV/borderline-IV hierarchy. The implication is substantive: the missing object is not more generic news text, but richer and cleaner pre-event expectation data, especially cloud-segment consensus or richer StreetAccount-style guidance fields.

9.6 Expectation-Surprise Data Interface

The final shock-design layer creates an auditable import interface for the object that would most directly support stronger causal claims: pre-event capex and cloud expectation surprises. The current populated import joins WRDS/IBES company CPX guidance to LSEG/Eikon capex-consensus snapshots observed strictly before announcement time, adds LSEG broker-level segment estimates for cloud and data-center segments, and adds a small transcript-coded fallback sample where the numeric benchmark is already documented before the event. For the segment rows, the announced segment revenue actual is compared with the mean of the latest strictly pre-announcement broker estimates for segments such as Intelligent Cloud, AWS, Google Cloud, Data Center, Oracle cloud support, and selected AI/data-center semiconductor segments. The same interface can also accept structured StreetAccount guidance, timestamped analyst capex forecasts, and hand-coded

transcript surprises, but only when the benchmark expectation is documented before the event timestamp. Rows with post-event consensus timestamps are rejected from shock construction. A separate LSEG structured-guidance pull still saves company-guidance rows for audit; those rows are not imported unless they also have an auditable pre-event benchmark. The detailed candidate table is retained in the generated artifacts, while the paper reports the promotion gate.

Table 13: Expectation-surprise promotion gate. True pre-event expectation-surprise candidates remain weak; the capex/infrastructure-SPY row is included only as the strongest event-return comparator.

Candidate	Family	Candidate events	Robust F	Macro pred. R^2	Relevance gate	Validity gate	Promotion decision
LSEG/Eikon CPX surprise	expectation surprise	29	0.100	0.206	weak	pass or not worse than baseline	diagnostic or appendix only
StreetAccount/LSEG cloud revenue surprise	expectation surprise	79	2.277	0.276	weak	pass or not worse than baseline	diagnostic or appendix only
LSEG/Eikon CPX plus cloud surprise	expectation surprise	92	3.194	0.159	weak	pass or not worse than baseline	diagnostic or appendix only
Analyst/IBES pre-event CPX surprise	expectation surprise	29	0.666	0.325	weak	pass or not worse than baseline	diagnostic or appendix only
Hand-coded transcript CPX/cloud surprise	expectation surprise	8	3.485	0.217	weak	pass or not worse than baseline	diagnostic or appendix only
Capex/infrastructure, SPY	event return comparator	118	20.443	0.239	borderline	pass or not worse than baseline	stronger robustness; reduced-form and AR still lead
EPS guidance placebo surprise	expectation surprise	11	0.037	0.203	weak	pass or not worse than baseline	diagnostic or appendix only

This layer is useful because it closes the most obvious data objection while preventing overclaiming. It writes the vendor-guidance interface and populates 123 pre-event expectation-surprise rows: 113 LSEG/Eikon CPX or LSEG segment-revenue rows plus 10 transcript-coded fallback rows. The strongest available expectation candidate is Hand-coded transcript CPX/cloud surprise with robust $F = 3.485$, so the gate is weak. This does not replace the baseline. It instead makes the next empirical requirement explicit: stronger causal claims require broader capex and cloud consensus-surprise coverage that improves relevance and validity, not another estimator. The source audit narrows the remaining data constraint to additional OCI and hyperscaler capex consensus coverage, richer timestamped analyst capex histories, and transcript-coded capex/cloud surprises benchmarked to independent pre-event expectations.

9.7 2017–2022 Extension

The 2017–2022 WRDS TAQ extension is a separate robustness layer with extension-prefixed outputs. It has 212 event rows, 212 finite event shocks, no source mixing in the preferred WRDS TAQ panel, 6 LSEG-reviewed Tier 2 timestamps, and 14 public/newswire fallback Tier 2 timestamps. Under the hybrid timestamp rule, only the LSEG-reviewed Tier 2 rows enter the preferred extension; the public/newswire fallback rows are appendix-only. Its robust first-stage F -statistic is 4.611. The extension therefore solves the coverage problem but not the relevance problem. It is useful as a power and external-validity check, but it is not promoted to the main empirical sample because the first-stage gate fails.

This distinction is important because the extension looks strong on data coverage alone. The extension has complete finite shock coverage (212/212), no mixed-source event windows, no missing benchmark returns in the gate table, and at least ten AI-basket names per event. Those facts make the extension a useful check on whether the 2022–2026 sample is unusually short. They do not make it a better identification sample: the preferred extension first stage is 4.611, well below

the paper’s borderline threshold of 10, macro predictability is higher than in the current baseline (0.416), and Ljung–Box/ARCH diagnostics flag serial-correlation and volatility warnings. The public/newswire Tier 2 rows also do not satisfy the same timestamp-quality rule as the current baseline. The extension is therefore reported as appendix and robustness evidence rather than as the main sample.

Table 14: 2017–2022 extension gate. The WRDS TAQ extension has complete event-window coverage, but relevance and validity diagnostics do not improve on the current sample, so the extension remains appendix and robustness evidence.

Gate	Value	Decision
event count	212	pass
finite shocks	212/212	pass
source mixing	0	pass
benchmark missing	0	pass
minimum AI-basket coverage	10	documented
LSEG-reviewed Tier 2 rows	6	preferred extension only
public/newswire Tier 2 rows	14	appendix fallback
macro predictability R2	0.416	validity warning
Ljung-Box p-value, lag 6	0.010	serial-correlation warning
ARCH-LM p-value, lag 6	0.014	volatility warning
robust first-stage F	4.611	fail relevance
paper use	coverage passes; relevance fails; appendix/robustness only	appendix/robustness

9.8 Timing Sensitivity

The timing-sensitivity layer recomputes the shock for the three caveated current-sample Tier 2 timestamps: GPT-4, Stargate, and AI Diffusion rescission. Under the hybrid rule, the main baseline uses the earliest audited market-observable LSEG/Refinitiv tape, but pre-announcement, social-wire, and non-official sources are explicitly caveated and tested with public/official timing and drop-caveated variants. In practice, the timing changes leave the first-stage classification unchanged. Dropping all three caveated events also leaves the design in the same borderline-IV category.

Table 15: Timing sensitivity for caveated current-sample Tier 2 events. Alternate public or official timestamps, and a specification dropping all three caveated events, do not materially change the first-stage gate.

Variant	Events	Finite events	Missing benchmark	Robust F	Partial R2	Gate
GPT-4 public timestamp	135	135	0	14.994	0.139	borderline iv robustness
Stargate official timestamp	135	135	0	14.994	0.139	borderline iv robustness
AI Diffusion public timestamp	135	135	0	14.994	0.139	borderline iv robustness
All caveats public timestamps	135	135	0	14.994	0.139	borderline iv robustness
Drop caveated events	132	132	0	14.600	0.137	borderline iv robustness

9.9 Estimator and Model Robustness

The second robustness layer changes the estimator. Reduced-form LPs are the main object because they do not divide by a borderline first stage. LP-IV gives the normalized relative response. Recursive VAR LP-IV and SVAR-IV are useful cross-checks only under stronger invertibility and recoverability assumptions, so they are interpreted as supporting evidence rather than as replacements for the weak-IV robust LP results.

Additional appendix modules are also reported. The BVAR uses Giannone et al. (2015) shrinkage to reduce variance, but shrinkage does not solve instrument relevance. Sign restrictions report identified-set bounds rather than only Haar-prior medians. Heteroskedasticity identification uses event versus non-event variance shifts as a diagnostic module rather than as a stand-alone identifying design. FAVAR expands the information set to reduce non-fundamentality. Mixed-frequency state-space methods avoid naive interpolation of quarterly TFP and capex. Beveridge–Nelson decomposition separates permanent and transitory components of the estimated response. These extensions are valuable for interpretation, but the main claim remains anchored in the reduced-form LPs and Anderson–Rubin sets. Optional extension data from Treasury futures, LSEG news intensity, OptionMetrics, Databento trades, and real-side infrastructure series are best read as appendix or future-work channels unless they improve the pre-registered shock validity or first-stage relevance.

10 Conclusion

This paper builds a high-frequency AI-exposed earnings and infrastructure event-shock proxy and uses it to study monthly macro, financial, and firm-level responses. The empirical evidence is deliberately organized around what the diagnostics support. The baseline 2022–2026 sample is complete and reproducible, but the first stage is borderline rather than strong and is weaker once the event-window component is removed from the monthly normalization variable. The paper therefore treats reduced-form event-shock local projections as the main empirical result, Anderson–Rubin sets as the main weak-IV uncertainty statement for conditional causal interpretation, and conventional LP-IV point estimates as conditional relative impulse responses.

Claims and limitations. The main finding is reduced form: market-observed AI earnings, infrastructure, and frontier-model events predict dynamic responses in financial and macro outcomes, with the clearest evidence in equity returns and more mixed real-activity responses. The broad firm panel adds a second reduced-form result: more AI-exposed firms have a stronger capex-to-assets response after positive AI event shocks, although the firm-level interaction-IV first stage is weak under two-way clustered inference.

The conditional causal interpretation requires the external-instrument exclusion restriction and a relevant first stage. Because the first stage remains borderline, and because the mechanical-overlap sensitivity sharply weakens it, the evidence does not support precise structural IRFs from conven-

tional LP-IV standard errors alone. The hybrid timestamp rule keeps first market-observable tape in the 2022–2026 baseline while flagging pre-announcement, social-wire, and non-official sources through timing/drop sensitivity. The 2017–2022 extension passes coverage checks but fails the relevance and validity gates, and its public/newswire Tier 2 timestamps are appendix-only.

The guidance and expectation-surprise screens clarify what would be needed for a sharper external-instrument specification. Text-screened LSEG/IBES guidance, structured WRDS/IBES guidance, LSEG/Eikon capex consensus, LSEG segment expectations, and transcript-coded fall-back rows all move closer to the desired pre-event surprise object, but their first stages remain weak in the current sample. The remaining data need is therefore narrow: additional OCI and hyperscaler capex/cloud expectation coverage, richer timestamped analyst capex histories, or transcript-coded surprise evidence with independent pre-event benchmarks that improves relevance without worsening validity. BVAR, proxy-SVAR, frequency-domain, sign-restriction, and state-space exercises strengthen the diagnostic and interpretive case, but they do not by themselves create instrument relevance or justify FEVD and historical-decomposition claims without additional recoverability, scale, and full structural-decomposition assumptions.

A Additional Channels and Data Extensions

This appendix records optional channels collected during the project and states whether each one adds enough value to keep in the draft. The rule is the same as in the main text: an appendix channel is useful if it clarifies interpretation, validates measurement, or defines a credible future data path. It is not used to strengthen causal claims unless it improves the pre-registered shock validity or first-stage relevance.

A.1 Belief, News, and Uncertainty Channels

The fixed-income event windows are conceptually separate from the equity event-shock proxy. Treasury-futures reactions around pure model releases can measure belief updates about transformative AI in the spirit of Andrews and Farboodi (2025), but they should not be merged with the equity capex/infrastructure proxy without an explicit multi-instrument system that passes relevance, exclusion, and overidentification diagnostics.

The LSEG news-intensity measures are useful for event classification and for checking whether headline volume is driving results. They are not, by themselves, exogenous shocks. The tighter LSEG headline/story screen in Section 9 shows the value and limit of this channel: it isolates economically cleaner guidance-related windows but does not strengthen the first stage.

The OptionMetrics seed and Databento trade-level data are retained only as appendix material. Option data could support an uncertainty channel if expanded to the full event universe. Trade data support event-window measurement quality by showing that the strongest windows contain real transaction activity, but they do not address macro identification.

Table 16: Appendix-channel decision summary. These data products are retained only when they add interpretation, measurement validation, or a clear future path; none currently changes the main reduced-form/AR/conditional-LP-IV hierarchy.

Channel	Current evidence and value added	Paper decision
Treasury futures	Pure model-release windows from Databento futures; useful for a separate belief-channel check related to long-rate reactions.	Keep as appendix/future work.
LSEG news intensity	Headline and story-text screens for AI, capex, cloud, and infrastructure language; useful for event classification and guidance-screen design.	Keep only as support for shock construction.
OptionMetrics	Narrow implied-volatility event-window seed; useful for a future uncertainty and tail-risk channel.	Future work unless expanded.
Databento trades	Trade-level checks for strongest clean events; useful for validating that event-window returns reflect active trading.	Keep as measurement support.
Real-side data	BEA fixed assets, EIA electricity, and related public series; useful for a future mechanism paper on physical investment constraints.	Future work; not a current result.
Surprise import	LSEG/Eikon CPX rows, LSEG segment estimates, and transcript-coded fallback rows; closest current path to stronger causal claims, but first stages remain weak.	Keep as diagnostic appendix.

A.2 Real-Side and Expectation-Surprise Channels

BEA fixed-assets, EIA electricity, and related infrastructure series are natural inputs for a mechanism paper linking AI news to data-center investment and energy demand. They are not a main result here because that design would need to map event shocks to real-side quantities at the correct frequency and separate AI-specific investment from broader technology and energy-demand movements.

The expectation-surprise import is the most valuable appendix channel because it targets the object needed for stronger causal claims: strictly pre-event capex and cloud expectation surprises. The populated import contains 123 accepted rows and rejects post-event consensus leakage, but its best available candidate remains weak. The appendix therefore keeps this layer as diagnostic evidence and as a precise data requirement for future work: broader OCI and hyperscaler capex/cloud consensus coverage, richer timestamped analyst capex histories, or transcript-coded surprises benchmarked to independent pre-event expectations.

B Methodology Verification

This appendix distills the project’s methodology and claims audit. Its purpose is to make clear which claims are supported by the implemented evidence and which claims require stronger assumptions or better data.

Table 17: Claim-by-claim methodology verification. The table separates identified objects from robustness, interpretation, and future-work objects.

Claim	Implemented evidence	Course or paper basis	Allowed interpretation
Event shock is a proxy	Complete 135-event baseline; $z_e = r_{B,e} - \hat{\beta}r_{M,e}$	High-frequency proxy logic in Causal Macro 3	Market-observed AI news proxy, not the structural shock itself
LP-IV object is relative	LP-IV estimates divide reduced form by the first stage	Proxy-SVMA/LP-IV normalization in Causal Macro 3	Relative IRF normalized by n_t , not a one-s.d. structural shock
Weak-IV caution binds	Baseline robust $F = 14.994$ and AR sets reported	Weak-IV warnings and Anderson–Rubin logic	Reduced-form and AR sets lead; point LP-IV is conditional
Reduced-form LPs are first-class	Main tables report responses to z_t before LP-IV	Local projection logic and lag-augmented LP slides	Predictive event-shock responses without first-stage division
LP-IV normalization is partly mechanical	Event-window-excluded n_t first stage is weak in the sensitivity table	External-instrument relevance and scaling logic	LP-IV is conditional rescaling; reduced form remains headline evidence
Lag-augmented inference has conditions	Main LP uses six lags and EHW/HAC sensitivity	LP inference slides; frequentist weak-dependence notes	Valid under stated persistence and regularity conditions, not arbitrary breaks
Spectral evidence is diagnostic	Coherency and band-share outputs are reported	Spectral 2 periodogram/coherency results	Frequency-domain association, not conventional FEVD
Advanced models are robustness	BVAR, proxy-SVAR, sign restrictions, and state-space outputs exist	GLP, Causal Macro 4, State Space notes	Interpretation and triangulation, not a substitute for instrument relevance
Extension is appendix evidence	2017–2022 extension has 212/212 finite shocks but $F = 4.611$	Same relevance gate as baseline	Coverage passes; relevance fails; appendix/robustness only
Guidance-surprise screens are diagnostic	Text/revision, structured IBES, LSEG segment-consensus, and transcript-coded expectation candidates remain weak	Pre-event expectation-surprise logic from external-instrument design	Stronger causal claims need broader capex/cloud consensus-surprise data that improves relevance

The central restriction is therefore simple. The paper can claim that AI-exposed earnings and infrastructure event-shock proxies predict later macro, financial, and firm-exposure responses. It cannot claim precise structural transmission from conventional LP-IV standard errors alone unless a future pre-registered shock clears the relevance and validity gates. Conventional FEVD and historical decomposition language is avoided unless recoverability, scale, and full structural-decomposition assumptions are imposed explicitly.

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